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Original article

In vivo hepatoprotective activity and the underlying mechanism of chebulinic acid from *Terminalia chebula* fruit

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ABSTRACT

Background: The fruit of *Terminalia chebula* Retz. is one of the most widely used herbal drug in Traditional medicine prescriptions including those for liver diseases. In the screening of bioactive constituents that have potential hepatoprotective activity, chebulinic acid (CA) which is a major chemical constituent of *T. chebula* fruit showed potent activity.
Purpose: This work was conducted to investigate the hepatoprotective activity and mechanisms of CA.
Methods: The hepatoprotective effect of CA was examined on hepatotoxic models of cells, zebrafish larvae and mice caused by tert-butyl hydrogen peroxide (t-BHP), acetaminophen (APAP) and CCl_4, respectively.
Results: Pretreatment with CA could prevent t-BHP-induced damage in L-02 hepatocytes by blocking the production of ROS, reducing LDH levels and enhancing HO-1 and NQO1 expression via MAPK/Nrf2 signaling pathway. In animal experiments, CA significantly protected mice from CCl_4-induced liver injury, as demonstrated by reduced ALT, AST and MDA levels, enhanced SOD activity, improved liver histopathological changes, and the activation of the Nrf2/HO-1 signaling pathway. CA metabolized to chebulic acid isomers with DPPH radical scavenging activity. In a transgenic zebrafish line with liver specific expression of DsRed RFP, CA diminished the hepatotoxicity induced by 10 mM APAP.
Conclusion: Experiments in cell and two animal models demonstrated consistent results and comprehensively expounded the hepatoprotective effects of CA.

Introduction

The liver is the primary organ responsible for detoxification and material metabolism and represents a central link in the development of oxidative stress (Li et al., 2019). Many risk factors, such as drugs, irradiation, alcohol, and environmental pollutants could induce oxidant stress in liver (Li et al., 2015). Previous studies have shown that oxidative stress can induce many types of hepatic diseases, such as fatty liver (Zhang et al., 2011), liver fibrosis (Yamamoto et al., 2016), and liver cancer (Wang et al., 2016).

The fruit of *Terminalia chebula* Retz. has a long history of medicinal use for intestinal and hepatic detoxification as well as respiratory conditions (Chinese Pharmacopoeia Committee, 1998, 2020; Gao et al., 2016). *T. chebula* fruit is an essential component of Mongolian compound medicine Gurigumu-7 which is an important traditional medicine used for liver diseases (Chinese Pharmacopoeia Committee, 1998;Xu et al., 2016). *T. chebula* fruit contains a variety of chemical components including tannins, phenolic acids, and flavonoids (Nigam et al., 2020). *Terminalia* phenolics are regarded as the bioactive constituents of *T. chebula* fruit and Gurigumu-7 (Upadhyay et al., 2014; Ajala et al., 2014; Xu et al., 2016). Chebulinic acid (CA), a phenolic compound found in *T. chebula* fruit at around 25 mg/g (Juang et al., 2004), has a

Abbreviations: APAP, Acetaminophen; ALT, Alanine transaminase; AST, Aspartic transaminase; CA, Chebulinic acid; DCFH-DA, Dichlorofluorescein diacetate; DMSO, Dimethyl sulfoxide; HO-1, Heme oxygenase-1; HPLC, High performance liquid chromatography; LDH, Lactate dehydrogenase; MDA, Malondialdehyde; MS, Mass spectrometer; MTT, 3-[4,5-Dimethylthiazol-2yl]-2,5-diphenyltetrazolium bromide; NMR, Nuclear magnetic resonance spectrometer; NQO1, NAD(P)H quinone oxidoreductase-1; Nrf2, Nuclear factor erythroid 2-related factor 2; PBS, Phosphate buffered saline; ROS, Reactive oxygen species; SOD, Superoxide dismutase; t-BHP, Tert-butyl hydrogen peroxide.

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wide spectrum of bioactivities such as antioxidant, anticancer, and antihypertensive (Munawar et al., 2019). CA was found to be able to distribute to liver (Lu et al., 2019) and contribute to the hepatoprotective effect of *T. chebula* water extract (Nigam et al., 2020), indicating that CA might be an effective hepatoprotective component. To confirm the hepatoprotective effect and investigate the underlaying mechanisms of CA, the present research was carried out using three hepatic injury models.

Tert-butyl hydroperoxide (t-BHP), CCl_4 and APAP which are all metabolized in liver, are used in cellular and animal models to induce oxidative damage (Feng et al., 2018; Lee et al., 2018; North et al., 2010).

Nuclear factor erythroid-2-related factor 2 (Nrf2) which is highly expressed in the liver plays a critical role in regulating a series of phase II detoxification enzymes and non-enzymatic antioxidants (Feng et al., 2018; Tang et al., 2014). Normally, Nrf2 binds to kelch-like ECH-associated protein 1 (Keap1) in the cytoplasm. Under oxidative stress, the inactivation of Keap1 causes Nrf2 to dissociate from Nrf2-Keap1 complex and translocate into the nucleus, where it combines with the antioxidant responsive element (ARE) and mediates the expression of genes with antioxidant and detoxifying activities, such as heme oxygenase-1 (HO-1) and NAD(P)H quinone oxidoreductase-1 (NQO1) (Tang et al., 2014). Studies have also shown that members of mitogen-activated protein kinase (MAPKs) family play a key role in the nuclear translocation of Nrf2 (Feng et al., 2018).

In the present study, we investigated the protective effects and the underlying mechanisms of CA against liver damage induced by t-BHP, APAP and CCl_4 in cell, zebrafish larvae and mice, respectively.

Materials and methods

Chemicals and reagent

Solvents for extraction and isolation were of analytical grade and purchased from XiLong Chemical Co. Ltd., Guangdong, China. L-02 cells were obtained from Procell Life Science & Technology Co., Ltd (Wuhan, China). Tg(fabp10a: dsRed; ela3l: EGFP) transgenic zebrafish larvae were obtained from China Zebrafish Resource Center (Wuhan, China). Silymarin, 3-(4, 5-dimethylthiazol-2-yl)-2, 5-diphenyltetrazolium bromide (MTT), t-BHP, dichlorofluorescein diacetate (DCFH-DA) were obtained from Sigma-Aldrich (St. Louis, Mo, USA). APAP was obtained from Tokyo Chemical Industry (Shanghai, China). Extracellular regulated kinase (ERK) inhibitor PD98059, p38 MAP kinase (p38) inhibitor SB203580, c-Jun N-terminal kinase (JNK) inhibitor SP600125, and NQO1 inhibitor dicoumarol were obtained from MedChem Express (Shanghai, China). Zinc protoporphyrin IX (Znpp, HO-1 inhibitor) was obtained from Cayman (Ann Arbor, MI), and brusatol (Nrf2 inhibitor) from Yuanye Bio-Technology (Shanghai, China). Primary antibody's name, source, dilution ratio, and manufacturer are listed in Table 1.

Table 1
The of primary antibodies.

Name of antibody	Source	Dilution ratio	Company
Nrf2	The rabbit	1:1000	Proteintech (Wuhan, China)
HO-1	The rabbit	1:1000	Proteintech (Wuhan, China)
NQO-1	The rabbit	1:1000	Proteintech (Wuhan, China)
ERK	The rabbit	1:1000	Abclonal (Wuhan, China)
JNK	The rabbit	1:1000	Abclonal (Wuhan, China)
p38	The rabbit	1:1000	Abclonal (Wuhan, China)
phospho-ERK	The rabbit	1:1000	Abclonal (Wuhan, China)
phospho-JNK	The rabbit	1:1000	Abclonal (Wuhan, China)
phospho-P38	The rabbit	1:1000	Abclonal (Wuhan, China)
Lamin B	The rabbit	1:1000	Wanlei (Shenyang, China)
GAPDH	The rabbit	1:1000	Wanlei (Shenyang, China)

Isolation of CA from T. chebula fruits

The dried fruits of *T. chebula* produced in Guang Xi China were collected by Hebei Renxin Yaoye Co., Ltd in July 30, 2016 (Lot number, 45216002, Production license: Ji20150012) and sold by Jv Yuan Tang pharmacy store (Hohhot, China) in December, 2018. The fruits were confirmed by the authors as the dried fruits of *T. chebula* through morphological observation and UPLC-MS analysis. The voucher specimen (NPFFT-3) is stored in the laboratory of natural products and functional foods, school of life science, Inner Mongolia University, China.

T. chebula fruits (2 kg) were knocked to pieces and extracted with methanol under ultrasound for three times. The extract solution was concentrated under reduced pressure and suspended in water followed by extracting with petroleum ether, ethyl acetate and n-butanol successively. Using high-performance liquid chromatography-mass spectrometry (UPLC-MS) (Agilent Technologies Singapore (International) Pte. Ltd., Singapore) analysis, the ethyl acetate soluble part was found to contain more CA than the other parts. A portion of the ethyl acetate soluble part (100.57 g) was applied to an ODS (Wako Pure Chemical Industries, Ltd., Osaka, Japan) column and the 20%-30% methanol eluate was collected. After vacuum concentration, a precipitate appeared. The precipitate was collected by centrifugation to obtain a white powder (468.02 mg after dried) which was identified as CA (purity > 95%) using ^{1}HNMR and liquid chromatography- mass spectrometry (UPLC-MS) analysis.

DPPH radical scavenging ability assay

DPPH (Sigma-Aldrich, St. Louis, Mo, USA) radical scavenging assay was carried out referring to the previous method (Xu et al., 2018).

Cell culture, cell viability, LDH release determination and Hoechst 33342 staining

L-02 cells were grown in RPMI 1640 medium (Thermo Fisher Scientific, Inc. Shanghai, China) supplemented with 10% fetal bovine serum (Kibbutz Beit Haemek, Israel) containing 100 U/ml penicillin and 100 μg/ml streptomycin. The cells were cultured at 37 °C in 5% CO_2 humidified atmosphere.

L-02 cells were seeded in 96-well plates at a density of 5×10^3 cells or in 6-well plate at a density of 2×10^5 cells per well and cultured for 24 h at 37 °C. The cells were treated with various concentrations of CA (0, 6.5, 13, and 26 μM) for 24 h, and t-BHP (0, 200, 400, 600, 800, and 1000 μM) for 3 h, respectively. After the treatments, MTT assay was carried out using previously described method (Liang et al., 2018) with chebulic acid (10 μM) as the positive control (Jung et al., 2019). The absorbance was read using a microplate reader (Tecan, Switzerland) at 570 nm. L-02 cells (5×10^3) were seeded in 96-well plates. After attachment, the cells were pretreated with or without Znpp, dicoumarol, brusatol, PD98059, SP600125 or SB203580 (20 μM) for 1 h and then incubated with or without 26 μM CA for 24 h in the absence or presence of 600 μM t-BHP for further 3 h. The assay was designed according to a previously reported method (Feng et al., 2018). Subsequently, MTT assay was carried out.

Determination of lactate dehydrogenase (LDH) release was performed using an LDH assay kit (Nanjing Jiancheng Bioengineering Institute, Nanjing, China) following the manufacturer's instructions. Hoechst 33342 staining was carried out according to the manufacturer's protocol (Wanlei, China).

Determination of ROS

ROS was detected by the nonfluorescent probe DCFH-DA referring to the previous method (Xu et al., 2018; Feng et al., 2018).

Immunofluorescence assay

L-02 cells were seeded in a 6-well plate at a density of 2×10^5 cells per well. After attachment, the cells were treated with CA (26 μM) for 24 h. Immunofluorescence analysis was carried out referring to a previous method (Liang et al., 2018).

Total RNA extraction and real-time RT-PCR

Total RNA was extracted using Trizol reagent (Takara, China) after L-02 cells were treated with CA. The RNA content and quality were assessed with the optical density at 260 nm and 280 nm, using NanoDrop 2000 (Wilmington, USA). cDNA was synthesized using a PrimeScript RT reagent kit with a gDNA Eraser (Takara, Japan), following the experimental procedure in the instruction manual. Amplification of cDNA was carried out according to the protocol of the assay kit (SYBR Premix Ex Taq IIkit, Takara, Japan). The primers used were designed by the software primer 5 and synthesized as listed in Table 2.

Western blot

The whole protein was harvested after L-02 cells were treated with CA using complete protein extraction kit (keyGEN BioTECH, jiangsu, China), so does the total proteins of liver tissue. The nuclear and cytosolic proteins were extracted using a nuclear protein and plasma protein extraction kit (Phygene, Fuzhou, China). A bicinchoninic acid (BCA) protein quantitative kit (Phygene, Fuzhou, China) was used to detect the protein concentrations. Western blot assay was carried out referring to the previous method (Xu et al., 2018).

Analysis of the intracellular metabolites of CA and the antioxidant activity of the cell extract

The experiment was carried out based on a method previously reported (Meng et al., 2015) with modification. Briefly, L-02 cells were seeded in 60 mm dishes and cultured overnight, then incubated with or without CA (26 μM) for 2, 4, 6, 12, and 24 h at 37 °C at 5% CO_2. After incubation, the medium was removed, and then the cells were washed with PBS three times and dispersed by trypsin and collected in 1.5 ml tubes by centrifugation. The collected cells were suspended in 200 μl methanol, and disrupted using an ultrasonic crusher (Ningbo, China). After centrifugation at 13000 × g for 20 min, 10 μl of the supernatant was taken out to assess DPPH radical scavenging activity, and the remaining supernatant was filtered with a 0.22 μm filter for UPLC-MS analysis. The optimized UPLC-MS parameters for the quantification are shown in Table 3. The chromatographic mobile phase composed of water containing 0.1% formic acid as A and acetonitrile as B. The elution program was 0.0 min: 5% B, 1.0 min: 7% B, 2 min: 30% B, 2.01 min: 100% B, 4 min: 100% B, and post time: 3 min. The flow rate was 0.3 ml/min, and injection volume was 2 μl.

Assess the hepatoprotective effect in zebrafish larvae against acetaminophen injury

Tg (fabp10a : dsRed ; ela3l : EGFP) transgenic zebrafish larvae of 72 h postfertilization (hpf) were examined with a fluorescence microscopy and those with red fluorescence in liver were selected and randomly placed into four wells (15 larvae per well) in a six-well plate. The control well contained 5 ml fish fluid, the APAP well contained 10 mM APAP, and the CA well contained 26 μM CA + 10 mM APAP. The positive control well contained 25 μg/ml silymarin + 10 mM APAP. At 96 hpf, the liver development of each group was observed under fluorescent microscopy. This experiment was performed according to the Institutional Animal Care and Use Committee protocols of Inner Mongolia University.

Table 2
The primer sequences designed for RT-PCR.

Gene name	sequences	Primer sequence (5'-3')
GAPDH	sense	GGAGTCCACTGGCGTCTT
	antisense	TGAGTCCTTCCACGATACC
HO-1	sense	AAGACTGCGTTCCTGCTCAAC
	antisense	AAAGCCCTACAGCAACTGTCG
NQO1	sense	TGGTCGGCAGAAGAGCACT
	antisense	GTCTCGGCAGGATACTGAAAG

Table 3
Optimized UPLC-MS conditions for the compounds.

Compound	MW	Precursor ion	Product ion	Fragmentor	Collision energy
Chebulinic acid	956	955	275	325	45
Chebulic acid	356	355	337.1	100	10
Ellagic acid	302	301	283.8	150	30
Gallic acid	170	169	125	90	10

Experiments on mouse liver injury model induced by CCl_4

Animal experiments were carried out in accordance with ethical guidelines for the care of laboratory animals and approved by the animal experimental ethics review committee of Inner Mongolia University ([2019]004). Male ICR mice (6-8 weeks old, 25-30 g) were obtained from Beijing Vital River Laboratory Animal Technology (Beijing, China).

After one week of acclimation, the mice were randomly divided into seven groups (10 mice per group): the normal group, CCl_4 group, 20 mg/kg silymarin + CCl_4 group, CA (37.5, 75 and 150 mg/kg) + CCl_4 groups, and 150 mg/kg CA group (for toxicity test). CCl_4 was given by one intraperitoneal injection. Silymarin or CA was given by oral administration for 5 days. Animal treatment was carried out referring to the previous method (Xu et al., 2016).

Determination of transaminase activities in serum, as well as SOD activity and MDA level in liver tissue

The mouse blood samples were put standstill at room temperature for 1 h, followed by centrifugation at 13500 rpm for 5 min. The supernatant was measured for the ALT and AST level with AMS-300 automatic biochemical analyzer in Inner Mongolia university hospital. SOD activity and MDA level of liver were measured using assay kits from Nanjing Jiancheng Institute of Biotechnology, China and following the protocols.

Histopathology and immunohistochemical analysis

Histopathology and Nrf2 immunohistochemical analysis assays were carried out using previously described methods (Choi et al., 2015).

Statistical analysis

The data were presented as means ± SD. Statistical analysis was performed with a one-way analysis of variance (ANOVA) using Microsoft Office Excel 2010 (Microsoft Corporation, USA). Differences were considered being significant at $p < 0.05$.

Results

Isolation and identification of CA with DPPH radical scavenging activity

A white powder with *m/z* 955 in negative UPLC-MS was isolated from *T. chebula* fruit. UPLC-MS and NMR analysis confirmed that this compound was CA (Fig. 1A) with purity > 95%. CA showed strong DPPH radical scavenging activity with the half maximal effective concentration (EC_{50}) being 12.5 μM (Fig. 1B).

The protective effects of CA against hepatotoxicity induced by t-BHP

The cytotoxic effect of CA and t-BHP were evaluated with MTT method. After exposure to various concentrations of CA (6.5–26 μM) for 24 h, the results of cell viability assays demonstrated that the overall growth status for the CA-treated group were similar to those for the control group (Fig. 2A). Following treatment with different concentrations of t-BHP (0-1,000 μM) for 3 h, the viability of L-02 cells reduced significantly in a concentration-dependent manner. At 600 μM t-BHP, cell viability decreased to 50% (Fig. 2B). Subsequently, 600 μM t-BHP was selected as the concentration for further oxidative damage experiments. CA showed significant protective effect against hepatotoxicity induced by t-BHP ($p < 0.01$) and the effect was better than the positive control chebulic acid (Fig. 2C). To further confirm the protective effect of CA, LDH assay and Hoechst 33342 staining were carried out. The results showed that CA could significantly reduce LDH release ($p < 0.01$) (Fig. 2D) and cell death (Fig. 2E) compared with t-BHP group.

Following t-BHP treatment, intracellular ROS were generated and accumulated in large quantities. In the CA pretreated groups (6.5, 13, and 26 μM), the accumulation of ROS was inhibited (Fig. 3) compared with the t-BHP group.

Enhancement of HO-1 and NQO1 expression by CA in L-02 cells

HO-1 and NQO1 are important phase II detoxifying enzymes that play important roles during oxidative stress resistance. We investigated the effects of CA on HO-1 and NQO1 expression using western blot and RT-PCR analyses. After L-02 cells were treated with CA (0-26 μM) for 24 h, the expression of HO-1 and NQO1 increased at both the protein (13, 26 μM) and mRNA levels (26 μM) (Fig. 4A-C). ZnPP, a HO-1 inhibitor, and dicoumarol, an NQO1 inhibitor, significantly weakened the ability of CA to protect L-02 cells against oxidative damage induced by t-BHP (Fig. 4D-E).

CA promotes the expression of Nrf2

After treatment with different concentrations of CA for varying times, the L-02 cells increased expressions of Nrf2 in nucleus in a dose- (Fig. 5A) and time-dependent (Fig. 5B) manner, however, it was reversed by cytosol. The promotion of Nrf2 expression in nucleus by CA treatment was further confirmed by immunofluorescence assay (Fig. 5C). The cell viability in CA-treated group was reduced by the addition of Nrf2 inhibitor brusatol (Fig. 5D).

Effects of CA on the phosphorylation of MAPK signaling pathway components in L-02 cells

Incubation of L-02 cells with CA (6.5, 13, and 26 μM) for 24 h increased the phosphorylation levels of ERK, JNK and p38 MARK (Fig. 6A). The inhibitors PD98059, SP600125, and SB203580 weakened the ability of CA to protect L-02 cells from oxidative damage induced by t-BHP (Fig. 6B-D), and reduced the expression levels of Nrf2, HO-1, and NQO1 (Fig. 6E).

Antioxidant activities of the cytosolic CA and its intracellular metabolites

It has been reported that CA can be hydrolyzed to glucose, gallic acid and chebulic acid (Ding et al., 2001). Gallic acid was not detectable in the cell culture containing CA in the present experiment. Interestingly, the present experiment showed that CA was able to penetrate into L-02 cells and transformed into two chebulic acid isomers that had the same molecular weight as chebulic acid (Fig. 7A, C) but different retention times in the UPLC-MS analysis. The chebulic acid isomers were not detectable when incubating 26 μM CA in cell-free media in the same condition, indicating that the two isomers were metabolites but not hydrolysates of CA. The cytosolic solutions showed DPPH radical scavenging activity (Fig. 7B).

Effects of CA on serum ALT, and AST levels following CCl_4-induced liver injury in mice

As shown in Table 4, the levels of AST and ALT in the CCl_4 group were significantly higher than those in the control group. Whereas, compared with the CCl_4 group, the levels of ALT and AST in the CA-treated group decreased significantly. The protective effects of the medium (75 mg/kg) and high doses (150 mg/kg) of CA were even better than the positive control, silymarin (20 mg/kg). The transaminase levels in mice treated with 150 mg/kg CA were similar to those in the control group, indicating that CA had no toxicity on mice liver at the tested concentrations.

Effects of CA on the hepatic lipid peroxidation, SOD activity, liver histopathology and liver development

In animal experiment, a significant decrease of SOD activity and a striking increase of MDA level were observed in CCl_4 group compared with the normal group. CA and silymarin restored SOD activity ($p < 0.05$ or $p < 0.01$) (Fig. 8A) and reduced the MDA level compared with the CCl_4 group ($p < 0.01$) (Fig. 8B). By histopathological analysis, it was found that compared with normal group, the CCl_4 group showed severe liver injury with necrosis and inflammatory infiltration. CA- and

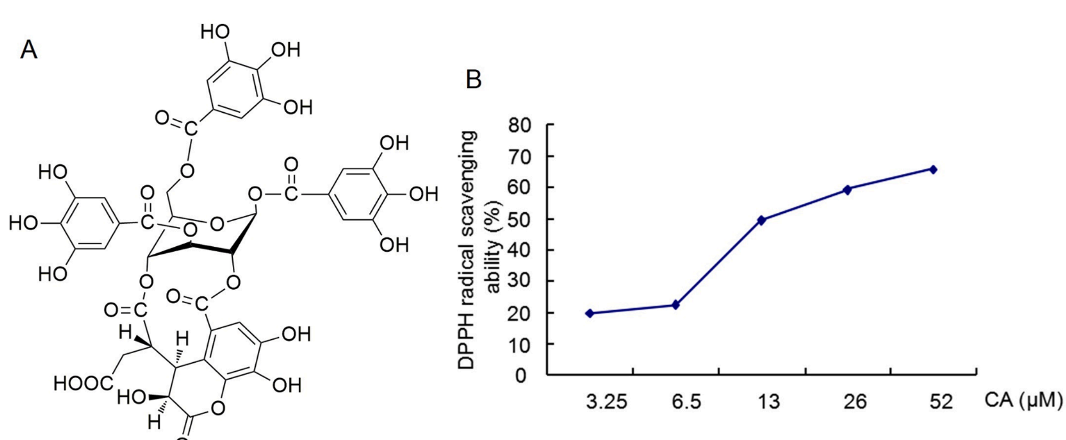

Fig. 1. Structure and DPPH radical scavenging activity of CA. (A) Chemical structure of CA. (B) DPPH free radical scavenging activity of CA.

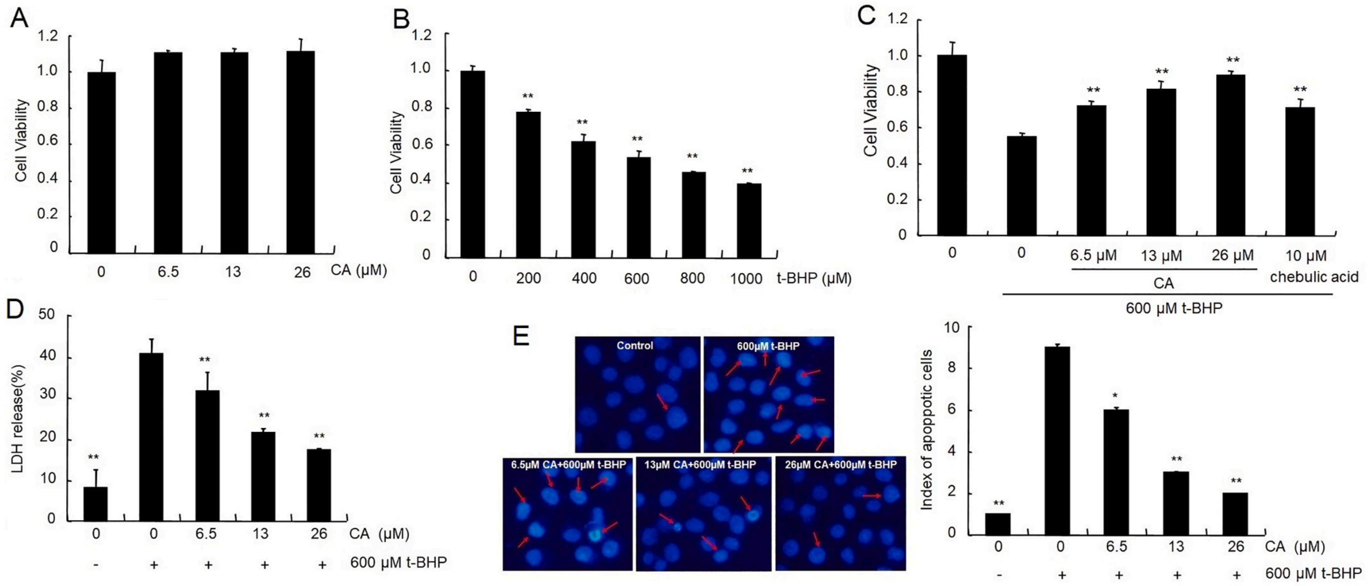

Fig. 2. CA resists oxidative damage induced by t-BHP in L-02 cells. (A) Cell viability following treatment with CA for 24 h as measured by MTT assay. (B) Cell viability following treatment with t-BHP for 3 h. (C) L-02 cell viability, (D) LDH release, and (E) L-02 cells stained with Hoechst 33342 and observed with a fluorescence microscope. ** $p < 0.01$ compared with control (B), * $p < 0.05$ compared with t-BHP ** $p < 0.01$ compared with t-BHP (C-E).

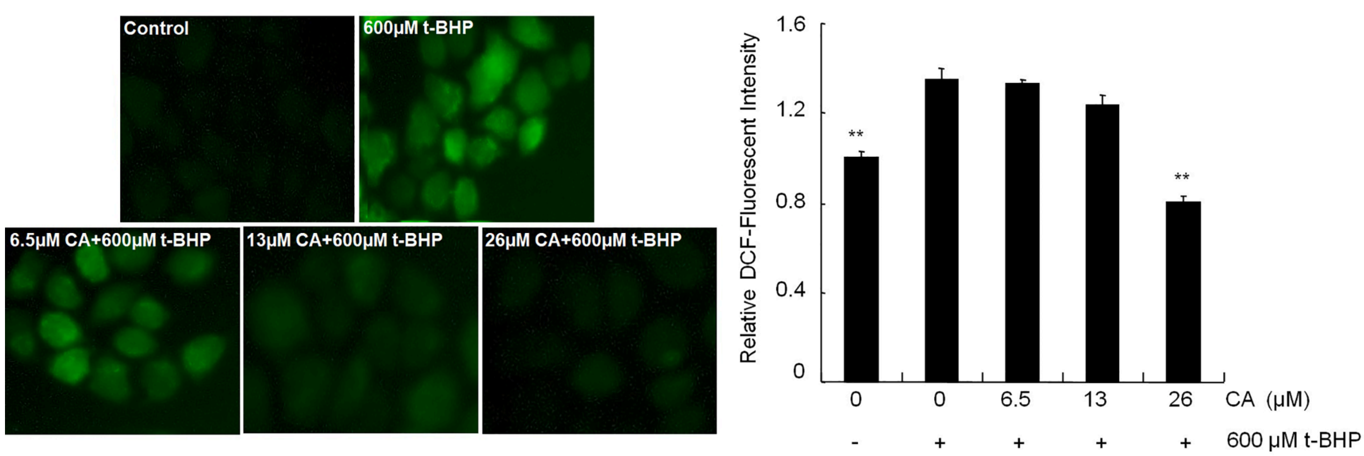

Fig. 3. CA inhibited the production of intracellular ROS induced by t-BHP. ** $p < 0.01$ compared with t-BHP.

silymarin-treated groups showed significantly reduced levels of liver injury, including mitigated necrosis and ameliorated histological changes (Fig. 8C). Using a transgenic zebrafish line Tg (fabp10a : dsRed ; ela3l : EGFP) with liver specific expression of DsRed RFP, the liver size of zebrafish larvae were examined. As shown in Fig. 8D, the liver in APAP-treated group was significantly smaller than those in control group. CA (26 μM) remarkably protected zebrafish larvae from APAP toxicity on liver development and the effect was better than the positive control silymarin. The liver in CA-treated group was much larger than those in APAP group.

CA protects against CCl_4-induced liver damage via Nrf2 signaling in mice

As shown in Fig. 9A, compared with the CCl_4 group, the groups that were pre-treated with a high dose (150 mg/kg) of CA or silymarin (20 mg/kg) upregulated Nrf2 expression. In addition, the CA pre-treated groups upregulated HO-1 expression. The immunohistochemical detection of Nrf2 is shown in Fig. 9B. Compared with the normal group, the expression of Nrf2 in the CCl_4 group increased. The expression levels of Nrf2 in 75 and 150 mg/kg CA ($P<0.05$) groups increased further compared with CCl_4 group.

Discussion

Application of natural antioxidants is a reasonable treatment strategy for liver diseases (Li et al., 2015). CA is a natural hydrolyzable tannin with good antioxidant activities (Manosroi et al., 2010). In this study, CA was found to significantly protect L-02 cells from t-BHP-induced oxidative damage by increasing the cell viability and scavenging intracellular ROS (Figs. 2 and 3).

Studies had indicated that Nrf2 can be activated by natural products and plays an effective role during the treatment of liver injury by promoting the expressions of HO-1 and NQO1 (Liang et al., 2018; Feng et al., 2018). In this study, CA increased the expressions of Nrf2, HO-1, and NQO1 (Figs. 4 and 5) in L-02 cells, and these results were confirmed in an animal experiment, which showed that CA upregulated Nrf2 and HO-1 expressions in the liver tissue of mice (Fig. 9).

Alternatively, MAPKs have been associated with the nuclear translocation and transcriptional activation of Nrf2 (Lee et al., 2015). In this study, following the treatment of L-02 cells with CA for 24 h, the levels

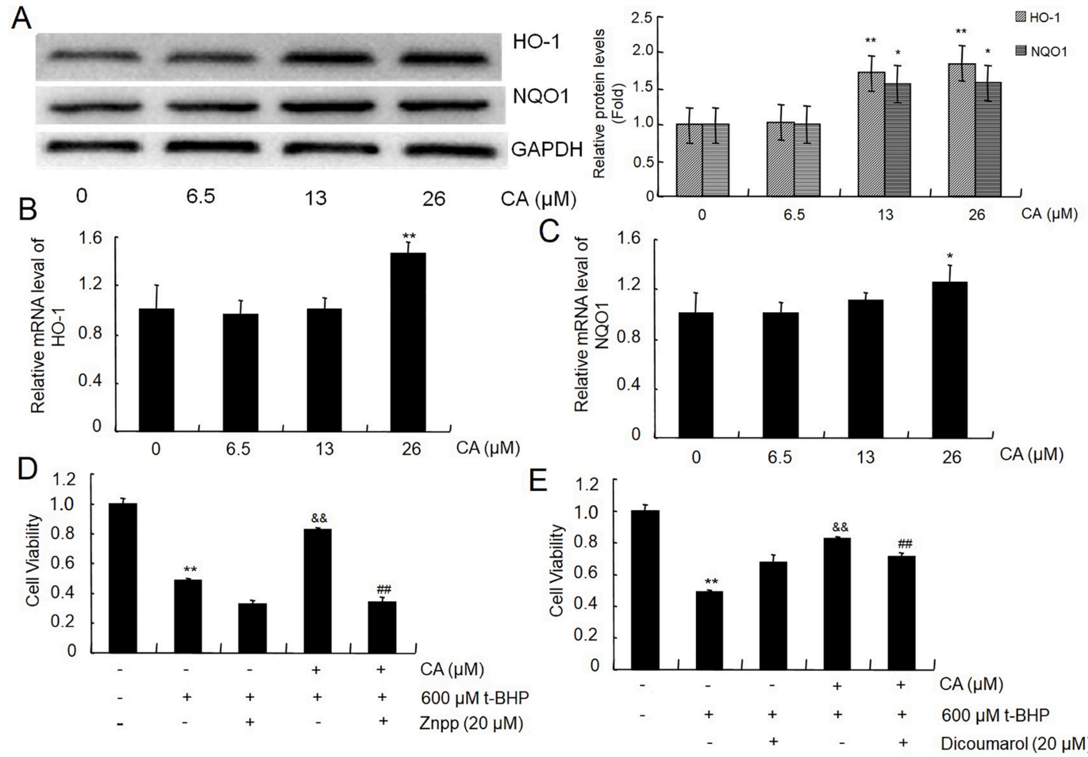

Fig. 4. Enhancement of HO-1 and NQO1 expression by CA in L-02 cells. L-02 cells were preincubated with CA (6.5, 13, and 26 μM) for 24 h, total protein and RNA were extracted and measured by western blot and RT-PCR, respectively. CA upregulated protein (A) and mRNA (B, C) expression of HO-1 and NQO1. The viability (D, E) of L-02 cells preincubated with HO-1 inhibitor Znpp and NQO1 inhibitor dicoumarol for 1 h prior to the treatment of CA (26 μM) for 24 h and t-BHP for 3 h. * $p < 0.05$, ** $p < 0.01$ compared with control, && $p < 0.01$ compared with t-BHP, ## $p < 0.01$ compared with the combination of t-BHP and CA (26 μM).

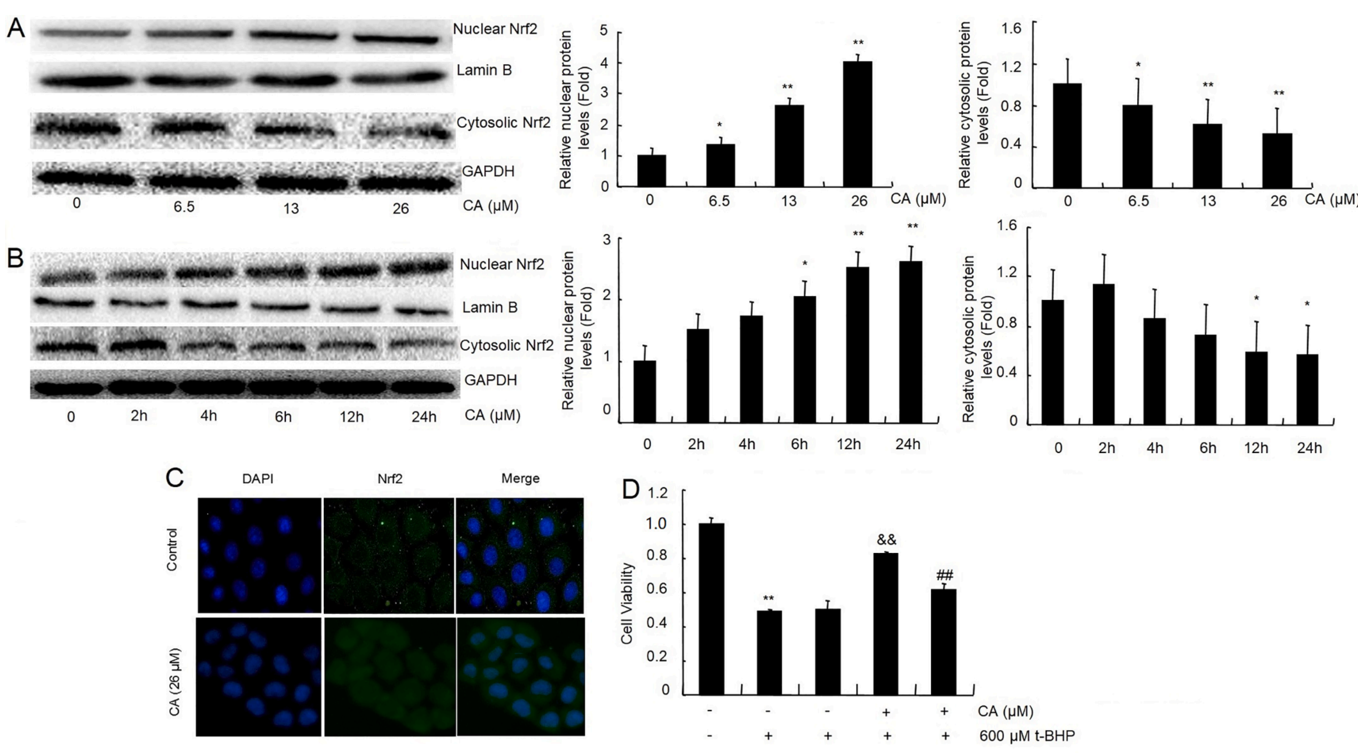

Fig. 5. Promotion of nuclear translocation of Nrf2 by CA in L-02 cells. L-02 cells were preincubated with CA (A) at 6.5, 13, and 26 μM for 24 h, or (B) at 2 h, 4 h, 6 h, 12 h, and 24 h for 26 μM. Nuclear and cytoplasmic protein levels of Nrf2 were analyzed by western blot. Lamin B and GAPDH were used as internal controls in nucleus and cytoplasmic, respectively. (C) Immunofluorescence of Nrf2 in L-02 cells. (D) L-02 cell viability. L-02 cells incubated with or without brusatol at 20 μM for 1 h and then treated with 26 μM CA for 24 h in the presence of t-BHP for 3 h. * $p < 0.05$, ** $p < 0.01$ compared with control, && $p < 0.01$ compared with t-BHP, ## $p < 0.01$ compared with the combination of t-BHP and CA (26 μM).

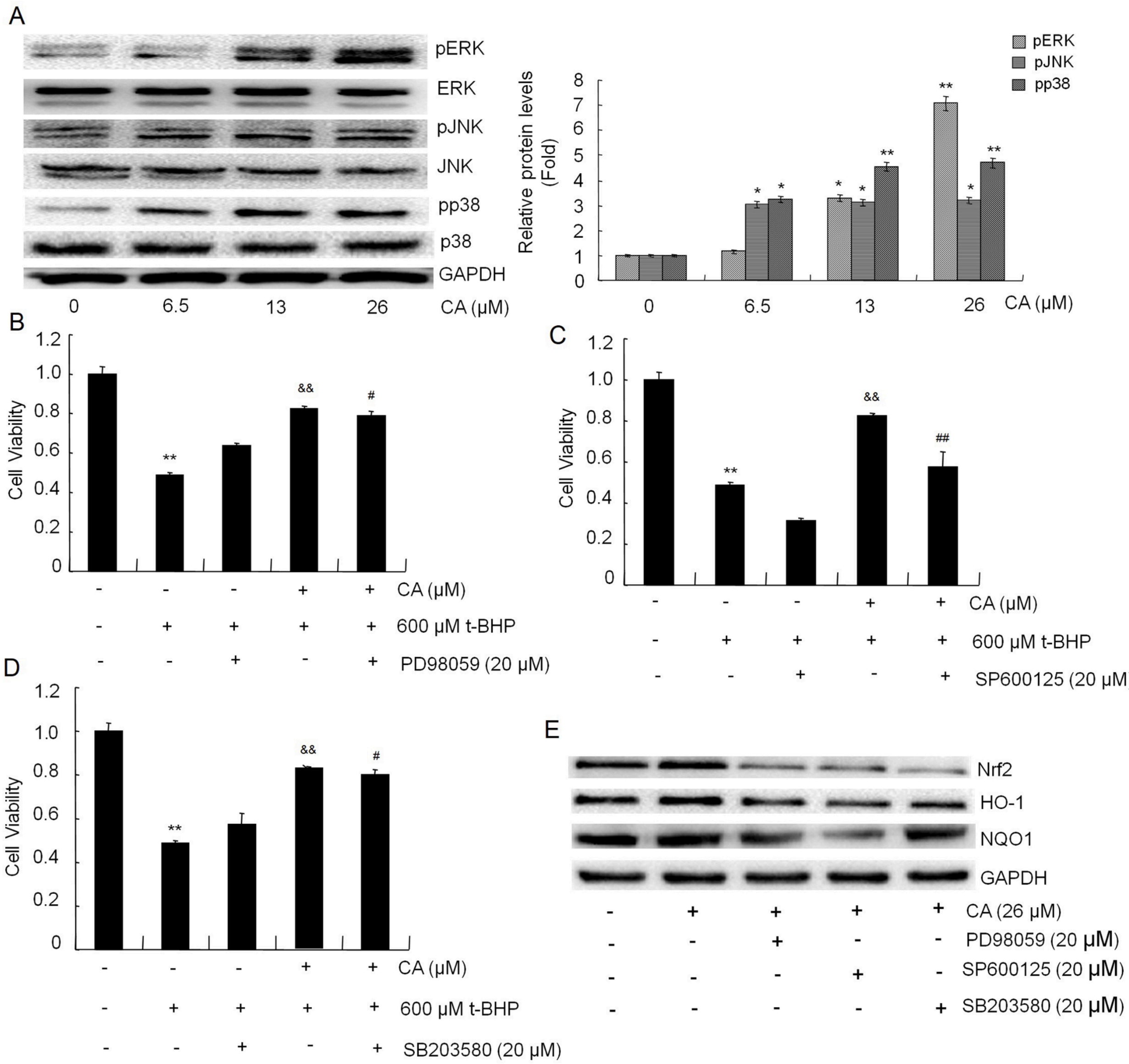

Fig. 6. Effects of CA on total and phosphorylated levels of the ERK, JNK and p38. (A) Protein levels of MAPK signaling pathway in L-02 cells preincubated with CA (6.5, 13, and 26 μM) for 24 h. (B, C, D) The viability of L-02 cells preincubated with 20 μM PD98059, SP600125, and SB203580 for 1 h and then treated with CA (26 μM) for 24 h in the presence of t-BHP for 3 h. (E) Protein levels of L-02 cells pretreated with 20 μM PD98059, SP600125, and SB203580 for 1 h, and then treated with 26 μM CA for 24 h. * $p < 0.05$, ** $p < 0.01$ compared with control, && $p < 0.01$ compared with t-BHP, # $p < 0.05$, ## $p < 0.01$ compared with the combination of t-BHP and CA (26 μM).

of pJNK, pERK, and pp38 were increased (Fig. 6). After the addition of inhibitors, the abilities of CA to protect L-02 cells from oxidative damage induced by t-BHP were weakened, these were similar to the previous report (Feng et al., 2018).

The hepatotoxicity mechanism induced by CCl_4 resembles that observed in human liver diseases (Taniguchi et al., 2004), which can cause elevation of serum transaminase levels, accumulation of MDA, depletion of SOD in tissue, and increased severity of histopathological liver damage (Yang et al., 2017). In this study, in the CCl_4 treated group, the serum levels of ALT and AST and the level of MDA in the liver significantly increased, whereas the SOD activity in the liver was significantly inhibited, resulting in serious liver histopathological changes, consistent with a previous study (Yang et al., 2017). However, pretreatment with CA mitigated these changes (Table 4, Fig. 8). The hepatoprotective effect of CA was further verified by the transgenic zebrafish larvae with red fluorescence in liver (Fig. 8D). Study has showed that a *T. chebula* extract containing CA could protect the liver against t-BHP induced oxidative damage (Choi et al., 2015). The present results confirmed that CA is a bioactive constituent of *T. chebula* fruit responsible for the hepatoprotective activity.

It is reported that when 100 mg/kg triphala in water containing 6.8 mg/kg CA was gavaged to mice, CA could be absorbed rapidly (Bei et al., 2013). The pharmacokinetic study showed that CA reached the peak concentration at 0.9 h after intraperitoneal injection (Lu et al., 2019). CA can be hydrolyzed into glucose, gallic acid, and chebulic acid (Ding et al., 2001). Study has shown that chebulic acid could protect HepG2 cells from t-BHP-induced oxidative stress through Nrf2 activation (Jung et al., 2019). Chebulic acid could also attenuate advanced glycation end products-mediated vascular dysfunction by inhibiting the production of ROS via the ERK/Nrf2 pathway (Nam et al., 2017). When 2000 mg/kg extract of *T. chebula* containing 29.4% chebulic acid was fed to rats for 14 days, no adverse effects were observed (Kim et al., 2012). In the present research, UPLC-MS analysis showed that CA penetrated cells and metabolized to chebulic acid isomers, which might act similarly as

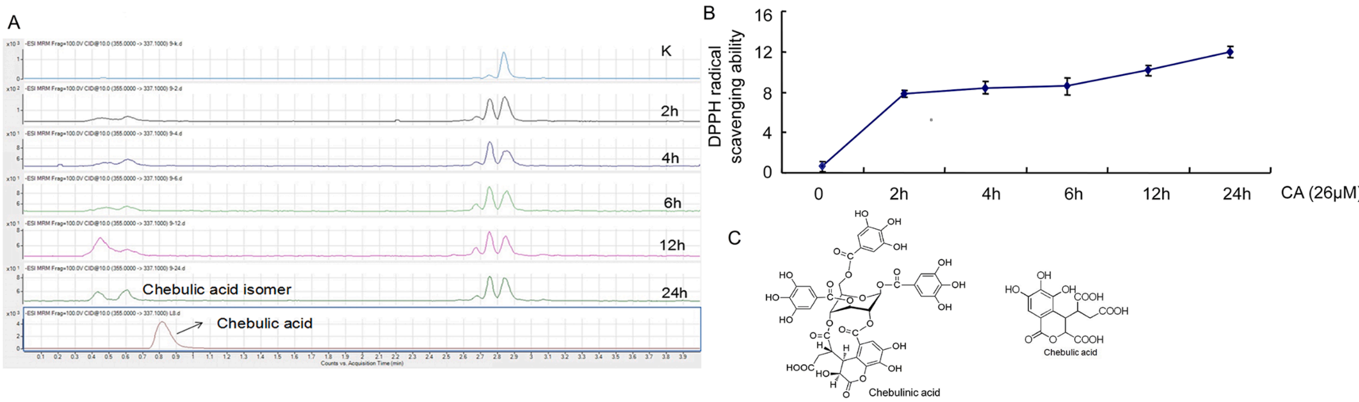

Fig. 7. CA penetrated L-02 cells and metabolized into chebulic acid isomers. (A) Chromatograms of cytosolic solutions and standard chebulic acid. After being treated with CA (26 μM) for 2, 4, 6, 12, and 24 h, the L-02 cells were collected and disrupted in 200 μl methanol under ultrasonication, and the supernatant was analyzed by HPLC-MS, (B) the antioxidant activity by DPPH radical scavenging assay. (C) Chemical structure of CA and chebulic acid.

Table 4
Effects of CA on serum ALT, AST in CCl_4-induced liver injury in mice.

group	transaminase (U/l) ALT	AST
normal	34.4±12.9**	90.6±12.2**
CCl_4	13204±1940.3	6515.3±1522.1
20 mg/kg silymarin+CCl_4	10244.4±3938.3*	4636.5±2207.1*
37.5mg/kg CA+CCl_4	12941.3±3297	5421.4±1049.8
75 mg/kg CA+CCl_4	9490±1916.1**	3121±460.2**
150 mg/kg CA+CCl_4	7928.6±1420.5**	2136±364.6**
150 mg/kg CA	34.8±6.7	109.7±15.6

Note: all values were expressed as means ±SD, * $p < 0.05$, ** $p < 0.01$, compared with CCl_4 group.

chebulic acid did (Jung et al., 2019) and play a key role in the *in vivo* bioactivity of CA.

Conclusion

The present study demonstrated that CA resisted the oxidative stress induced by t-BHP in L-02 cells though up-regulation of HO-1 and NQO1 expression levels, which was mediated by Nrf2 via the MAPK signaling pathway. Mice pretreated with CA significantly alleviated CCl_4-induced acute liver injury. Meanwhile, CA promoted the liver development in zebrafish larvae that were retarded by APAP. Our findings indicated that CA might be a promising leading compound for the development of therapeutic agents for liver injury.

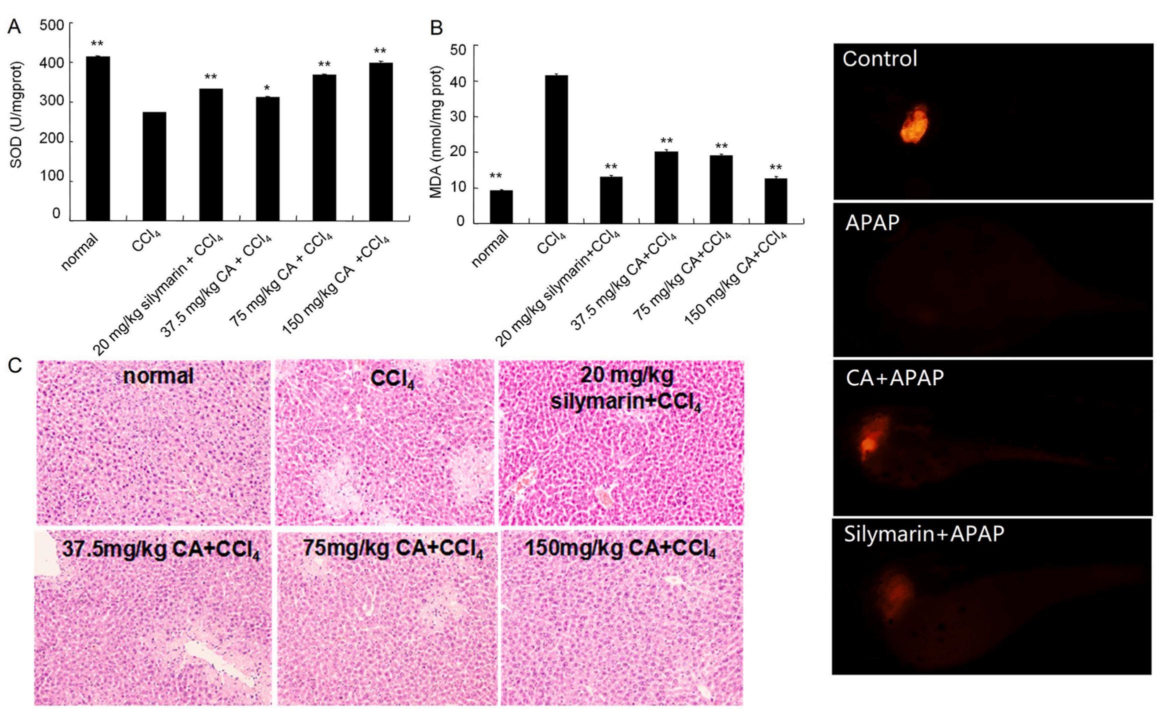

Fig. 8. Effects of CA on hepatic levels of MDA, SOD against CCl_4-induced damage and liver histopathology, and protection of zebrafish larvae from APAP-induced liver development toxicity. (A) The activity of SOD enzyme. (B) MDA levels in liver tissue. (C) Liver sections with the HE staining. Mice received intragastric administration of silymarin (20 mg/kg) or different doses of CA (37.5, 75, and 150 mg/kg) once daily for 5 consecutive days, and subsequently subjected to intraperitoneal single injection of CCl_4 (1%, v/v). (D) Liver expression of DsRed RFP in zebrafish larvae treated with indicated compounds. * $p < 0.05$, ** $p < 0.01$, compared with CCl_4 group.

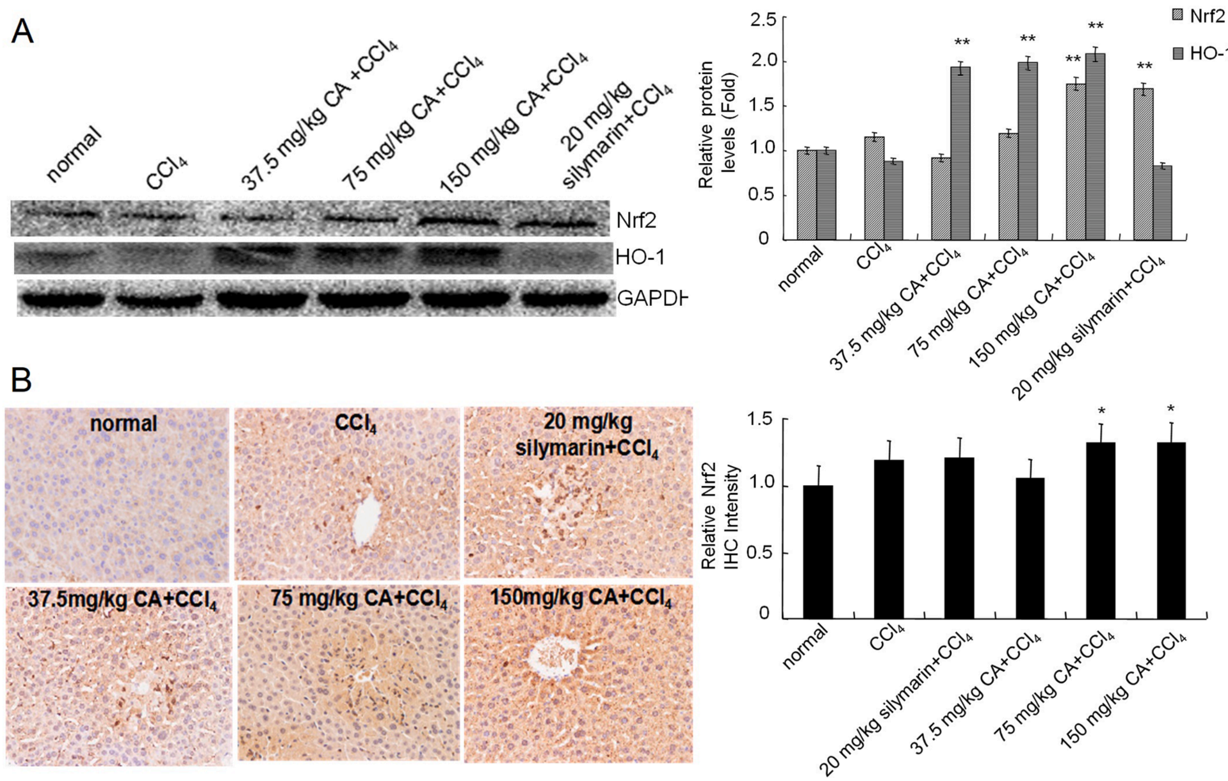

Fig. 9. Effects of CA on Nrf2 and HO-1 protein expression, and immunohistochemical detection of Nrf2. (A) Nrf2 and HO-1 protein expression. (B) the immunohistochemical detection of Nrf2. IPP6.0 software was used for optical density analysis of immunohistochemical photos. * $p < 0.05$, ** $p < 0.01$, compared with CCl_4 group.

CRediT authorship contribution statement

Xin-Hong Feng: Conceptualization, Methodology, Writing - review & editing. **Hai-Yan Xu:** Conceptualization, Methodology, Writing - review & editing. **Jian-Ye Wang:** Methodology. **Shen Duan:** Methodology. **Ying-Chun Wang:** Conceptualization, Supervision, Writing - review & editing. **Chao-Mei Ma:** Conceptualization, Supervision, Writing - review & editing.

Declaration of Competing Interest

The authors declare no conflict of interest.

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